

Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering

2022 – 2026

University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream)

2019

St. John's College, Panadura, Sri Lanka

Results: 3 A's (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Experience

R&D

Software Engineer — KōreroCare Platform

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland | Remote / Part-time

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern (Backend Focus)

Dec 2024 – May 2025

Information Systems Associates (Pvt) Ltd — Aviation Technology Company

- Engineered a distributed, high-throughput unique ID generator for Passenger Name Records (PNRs) using a Kafka-backed pool model, ensuring uniqueness and security across scaled instances.
- Optimized the PNR generation service to achieve ****25+ PNRs/sec throughput**** per node with ****<100ms latency****.
- Architected and implemented a hybrid integration strategy by embedding a gRPC client and Kafka consumer into a legacy Java 8 monolith.
- Investigated and refactored inefficient Kafka Streams processing logic, building a proof-of-concept to validate a more efficient Producer-Consumer model.
- Developed dynamic, database-driven feature toggles to control PNR routing logic.

Keywords: Java 18, Spring Boot, gRPC, Kafka, REST API, Couchbase, Oracle DB, PostgreSQL, Microservices, Distributed Systems.

Projects

See more projects on my portfolio

1. Employee Management System (EMS) (2025 | Individual)

Architecture | Github 

Designed and built a comprehensive, cloud-native microservices application for employee and department management, featuring a full-stack architecture with a React/TypeScript frontend and Spring Boot backend.

- Architected a distributed system using **Spring Cloud** for service discovery, centralized configuration, and an API gateway, with **Resilience4j** for robust fault tolerance.
- Enabled dynamic configuration management using **Spring Cloud Bus** with **RabbitMQ** and established a robust observability stack (Zipkin, ELK, Prometheus/Grafana).
- Designed and managed a CI/CD pipeline using **GitHub Actions**, with **ArgoCD** for GitOps-based deployments to the Kubernetes cluster.

Keywords: Java 17, Spring Boot, Spring Cloud, React, TypeScript, Resilience4j, RabbitMQ, Docker, Kubernetes, GitHub Actions, ArgoCD, Prometheus, Grafana

2. GoTogether (2024 | Group)

Architecture | Live  | Github 

Developed and deployed a full-stack travel assistant platform with a **Next.js** frontend (SSR/CSR, secure auth) and microservices in **Spring Boot** and **Go**, enabling AI itinerary generation via **Gemini API**.

- Implemented a **Spring Boot** social service with **Keycloak** integration for identity management and a dedicated **Go** service using **gRPC** for high-performance inter-service communication.
- Configured production-grade security with **NGINX**, **Certbot**, and **DuckDNS**, and orchestrated deployment on **Oracle Cloud Free Tier** using custom scripts and Docker optimizations.
- Built a dedicated **Auth Service using Keycloak**, managing OAuth2/OIDC flows, token issuance, and automated realm/client boot-time import scripts.

Keywords: Next.js, Go, Spring Boot, Keycloak, PostgreSQL, NGINX, Docker, Gemini API, Oracle Cloud, gRPC

3. Food Delivery Application (2024 | Group)

Github 

Built a scalable, event-driven food delivery platform inspired by systems like Uber Eats, using a modern microservices architecture. As part of the core team, I focused on designing and developing the Restaurant Service.

- Developed the Restaurant Module with secure sign-up/login using **Spring Security** and **JWT** (with extended claims).
- Built CRUD APIs for menu and item management; integrated **Redis caching** to improve performance and reduce database load, with appropriate cache invalidation strategies.
- Enabled order sync via **Kafka**: when an order is placed, an event is published to the 'order_created' topic, which the restaurant service consumes to update order details.

Keywords: Java, Spring Boot, gRPC, Kafka, Redis, PostgreSQL, WebSocket, Docker, React.js, Ant Design

4. Serverless Auction Platform (2025 | Individual)

Architecture | Github 

A capstone project built using AWS, following a microservices architecture with an event-driven, serverless (FaaS) design.

- Architected an event-driven system using **AWS Lambda**, **API Gateway**, and **EventBridge**, with **SQS** and **SNS** for asynchronous communication.
- Designed a high-performance **DynamoDB** data model with GSIs and implemented a custom **Lambda Authorizer** for JWT-based security.
- Utilized **LocalStack** and **Docker** for local cloud emulation and integrated **Middy** middleware for robust error handling.

Keywords: Node.js, Serverless Framework, AWS Lambda, DynamoDB, SQS, SNS, SES, JWT, Middy, LocalStack, Docker

5. Go gRPC GraphQL – Multi-Tenant Microservices (2025 | Individual)



Architected a **polyglot, multi-tenant microservices** platform with a focus on scalability, fault isolation, and unified data access.

- Developed backend services in **Go** using **gRPC** and **Protobuf** for efficient inter-service communication.
- Implemented a **GraphQL API Gateway** to provide a single, unified endpoint for client applications.
- Designed multi-tenant data storage with **PostgreSQL**, **Elasticsearch**, and optional **Cassandra** for high-volume workloads.

Keywords: Go, gRPC, GraphQL, Protobuf, PostgreSQL, Elasticsearch, Cassandra, Docker Compose

6. Fintech & Crypto Trading Automation Lab (Ongoing | Group)

Industry-Mentored Initiative (Ongoing)

- Selected for an exclusive initiative to build intelligent automation systems for fintech and crypto markets under industrial mentorship.
- Developing production-ready trading tools and gaining deep exposure to technical analysis (TA) and algorithmic trading strategies.
- Focusing on high-performance software engineering with strict deadlines and client-side expectations.

Keywords: Fintech, Crypto Trading, Forex, Algorithmic Trading, Technical Analysis, Intelligent Automation, Industrial Mentorship, Personal Finance & Wealth Management

7. Improving Blockchain Scalability: Throughput Enhancement in ZK-Rollups

Research Project — University of Moratuwa

- Built a modular ZK-rollup benchmarking framework to evaluate Ethereum Layer-2 trade-offs across sequencing, batching, proving, data availability, and L1 settlement.
- Implemented the Rust-based Submitter service for finalized batch submission, DA-mode selection, L1 bridge interaction, transaction receipt validation, and settlement metric export.

- Added calldata, EIP-4844-style blob, and off-chain / Validium-style benchmark modes, observing approximately 70% lower DA publication cost for blob-style mode compared with calldata.
- Introduced a mock-testable **BridgeClient** abstraction, improving Submitter testability and increasing verifiable test coverage from approximately 41% to 88%.
- Built Docker-based benchmark orchestration scripts, a Poisson-style workload generator supporting 8 TPS base and 80 TPS burst workloads, and Python data tools for throughput, P50/P95/P99 latency, finalization time, gas/cost, and DA-footprint analysis.
- Produced reproducible benchmark outputs including latency CDFs, cost heatmaps, and Pareto-style trade-off comparisons for the final research evaluation.

Technologies: Rust, Solidity, Hardhat, TypeScript, Python, Docker, gRPC, SQLite/PostgreSQL, RISC0 / Mock Proving Path, Matplotlib

8. Blockchain Supply Chain Management DApp (2025 | Individual) Architecture | Live | Github

Built a full-stack decentralized application (DApp) enabling end-to-end shipment tracking and secure escrow payments on the Ethereum blockchain (Polygon Amoy Testnet).

- Developed a production-grade Solidity smart contract with a strict 3-state lifecycle (PENDING → IN_TRANSIT → DELIVERED) and on-chain auditability.
- Implemented a trustless escrow mechanism that locks funds on creation and releases payments only upon verified delivery.
- Built a responsive Next.js/React frontend with global state management and Ethers.js integration for secure wallet connections.
- Designed a CI/CD pipeline using GitHub Actions to automate contract and frontend deployment, utilizing Hardhat for local simulation and verification.

Keywords: Solidity, Hardhat, Ethers.js, Web3Modal, Next.js, React Context API, Tailwind CSS, Polygon Amoy, Smart Contracts, Escrow Payments, CI/CD

Other Relevant Projects

9. Automated AI Shorts Video Creator (2025 | Individual)

An open-source project to automate the creation of short-form videos using a powerful combination of AI tools and workflow automation.

- Orchestrated a video creation pipeline using **n8n** as the central workflow engine.
- Integrated various AI services for script generation, text-to-speech, and content preparation, including **Together AI**, **OpenRouter**, and **ElevenLabs**.
- Containerized the entire stack using **Docker Compose** for one-click local deployment.

Keywords: n8n, Docker, AI, Workflow Automation, Video Generation

10. RPAL Interpreter – Functional Language Compiler (2024 | Group)

Engineered a complete compiler and interpreter for the RPAL (Right-reference Pedagogic Algorithmic Language), built from scratch to parse and evaluate programs according to official language grammar and semantics.

- Developed a Lexical Analyzer using custom Finite State Automata (FSA) and regex-based token identification.
- Built an Abstract Syntax Tree (AST) and Standard Tree (ST) generator.
- Implemented a CSE Machine to evaluate programs using 13 semantic rules of RPAL.
- Built a robust testing framework with **Pytest** covering over 168 test cases across AST, ST, and full output validation.

Keywords: Python, Pytest, GitHub Actions, Makefile, CLI

11. AI Blog Post Generator (2025 | Individual)

A multi-agent system that uses specialized AI agents to generate professional, well-structured, and well-researched blog posts.

- Built with the **Agno Framework** for its beginner-friendly approach to multi-agent workflows.
- Orchestrated research, content extraction, and writing using specialized agents for each task.
- Designed for provider flexibility, with easy switching between **OpenAI**, **Gemini**, **Claude**, and **Groq** models.

Keywords: Streamlit, Agno Framework, Pydantic

12. Maison Vella – E-Commerce Admin Panel (2024 | Individual) Architecture | Live | Github

Comprehensive admin interface for the Maison Vella e-commerce ecosystem, facilitating centralized management of products, orders, and content assets.

- Engineered a responsive **Next.js 14** app with **Server Actions**, **tRPC**, and **ShadCN UI**, implementing **CQRS** and layered architecture.

- Integrated **Sanity** (Headless CMS) via GraphQL and managed data with **Prisma** and **Supabase** in a serverless environment.
 - Secured access with **Clerk Auth** and optimized media delivery using **Cloudflare R2** and a global CDN.
- Keywords:* Next.js 14, Server Actions, tRPC, GraphQL, Prisma, Supabase, Sanity, Serverless, Cloudflare R2, CQRS

13. Content-Aware Meme Generator (2025 | Individual)

An automated system for generating contextually relevant memes from any URL, leveraging LLMs and the Memegen.link API.

- Orchestrated a multi-step meme generation workflow using **LangGraph**.
- Utilized **Llama 3.3** (via Groq API) for content analysis, concept generation, and text creation.

Keywords: Streamlit, LangGraph, Llama 3.3, Groq, Pydantic, Beautiful Soup

14. PaperMind – AI Research Assistant (2025 | Individual)

Developed a lightweight, **RAG-powered** platform for intelligent querying of academic papers.

- Designed a **LangChain** pipeline with **ChromaDB** vector storage for fast, context-aware lookups.
- Generated embeddings using **Sentence Transformers** for improved retrieval accuracy.
- Delivered LLM-generated responses using **Gemini API** and HuggingFace models.

Keywords: Streamlit, LangChain, ChromaDB, Sentence Transformers, Gemini API, HuggingFace

15. Virtual Me – 3D Animated AI Chatbot (2024 | Individual)

Created an interactive 3D virtual assistant with lifelike animations and AI-driven conversations.

- Integrated **Hugging Face** (Mixtral model) for natural language dialogue.
- Implemented **Eleven Labs** TTS for human-like voice output.
- Built immersive 3D interface with **React** and **Three.js**.

Keywords: React (Three.js), Node.js, Express.js, Hugging Face, Eleven Labs, Rhubarb, FFmpeg

Training & Certifications

CKA Training — KodeKloud, 2025 • AWS SAA-C03 Training — Udemy, 2025 • Apache Kafka Series — Udemy, 2025 • 100 Days of DevOps — KodeKloud, 2025 • Advanced OAuth Security — Udemy, 2024

Skills

Core	Java Spring Boot Microservices Distributed Systems System Design REST APIs
Backend	Spring Cloud Spring Security Resilience4j Node.js Express.js Gin
Messaging & APIs	Kafka RabbitMQ gRPC GraphQL WebSockets Protobuf JWT OAuth 2.0
Data	PostgreSQL MySQL Oracle DB Redis Couchbase
Cloud & DevOps	AWS Docker Kubernetes CI/CD (GitHub Actions, Jenkins)
Testing	JMeter Gatling Pytest Unit Testing
Tools	Git Gradle Postman Jira Confluence

Referees

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